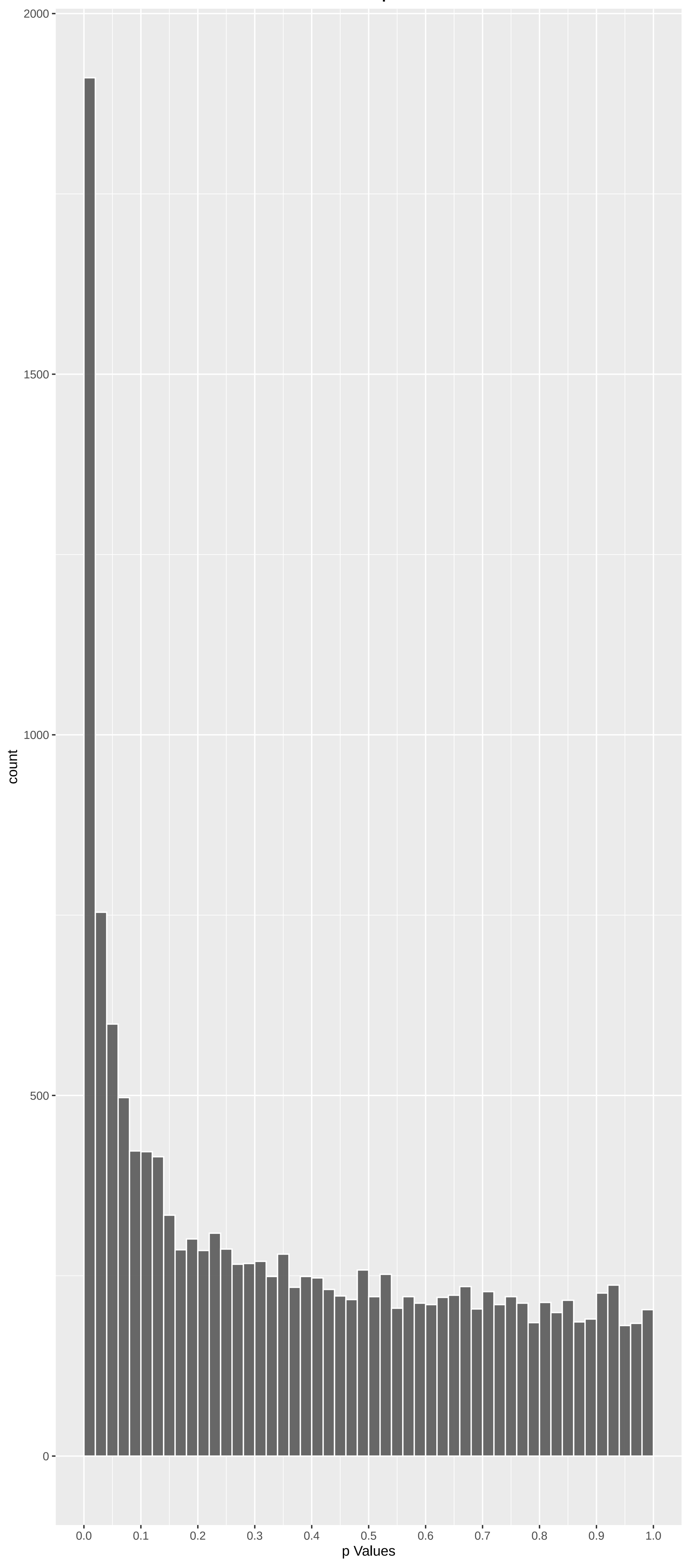
Top genes by P−value non−permulated

Gene	Rho	N	P	p.adj	qValueNoperm
LRIT1	0.11199011	346	5.105e−10	7.715e−06	1.166e−01
PDE6B	0.10626165	309	1.060e−08	7.449e−05	3.449e−01
TRPM6	0.08789351	366	1.479e−08	7.449e−05	3.449e−01
CPLX3	0.15258470	205	2.417e−08	9.130e−05	3.449e−01
CPB1	0.08431346	353	5.100e−08	1.541e−04	4.659e−01
GALNT5	0.07760340	354	1.659e−07	3.297e−04	5.028e−01
JAK3	0.09445217	290	1.745e−07	3.297e−04	5.028e−01
CFAP20DC	0.08094318	344	1.371e−07	3.297e−04	5.028e−01
TLE6	0.09539041	278	2.742e−07	4.363e−04	5.028e−01
AKNA	0.08200622	322	2.887e−07	4.363e−04	5.028e−01
CRYAA2	0.14840183	177	3.211e−07	4.411e−04	5.028e−01
TRPM1	0.07082851	362	4.268e−07	5.375e−04	5.028e−01
AFF1	0.07165443	353	5.109e−07	5.939e−04	5.028e−01
SMC4	0.07157356	350	5.795e−07	6.256e−04	5.028e−01
TASOR2	0.06873580	361	6.544e−07	6.391e−04	5.028e−01
ABCC10	0.06670428	371	6.766e−07	6.391e−04	5.028e−01
SPG11	0.06515057	377	7.444e−07	6.617e−04	5.028e−01
DHDH	0.09543525	253	9.388e−07	6.987e−04	5.028e−01
PHYH	0.07590208	317	9.709e−07	6.987e−04	5.028e−01
A2M	0.07001557	345	9.216e−07	6.987e−04	5.028e−01
THADA	0.06580886	369	8.604e−07	6.987e−04	5.028e−01
ACADM	0.07897548	299	1.227e−06	8.239e−04	5.208e−01
ADGRF1	0.06418727	367	1.254e−06	8.239e−04	5.208e−01
ADAMTS3	0.06708037	348	1.403e−06	8.833e−04	5.208e−01
MUC20	0.08246624	276	1.915e−06	1.082e−03	5.208e−01

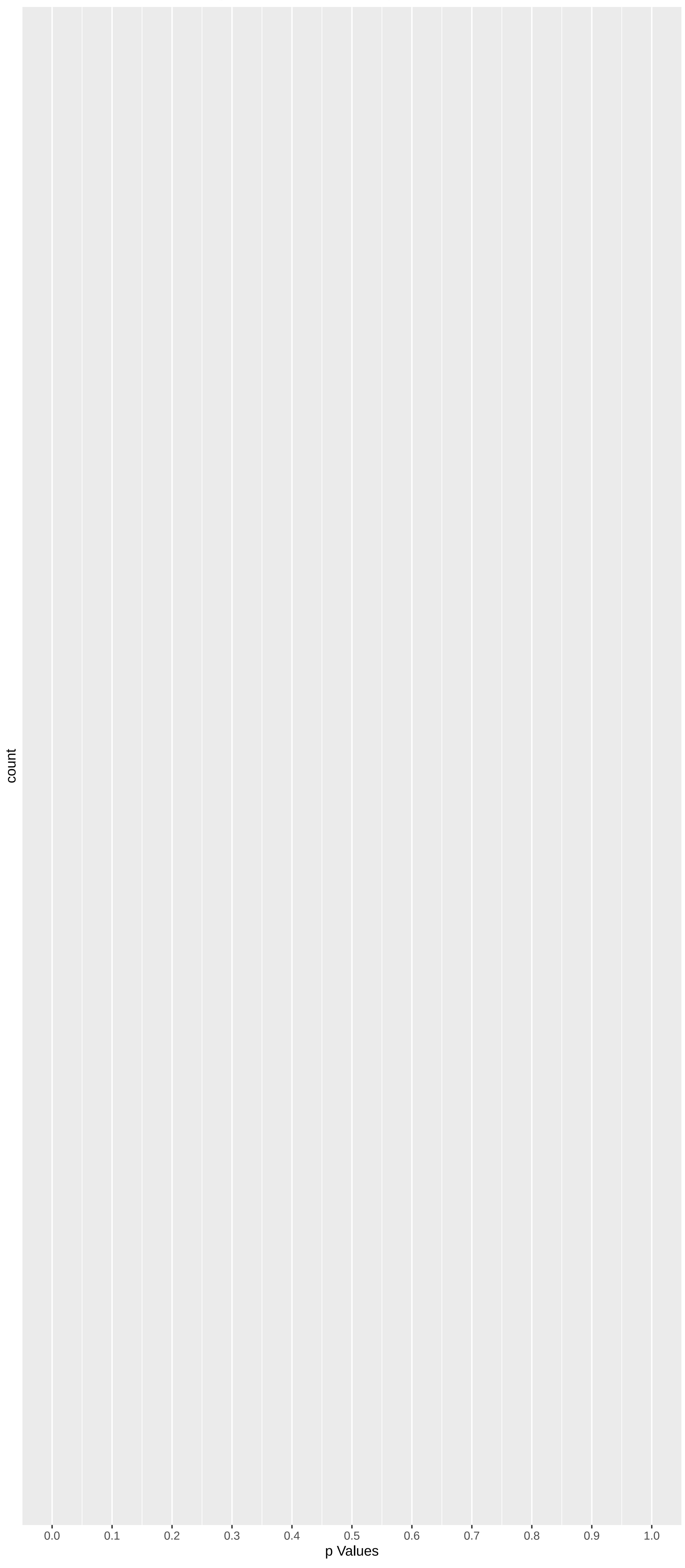
Top genes by Q−Value non−permulated

Gene	Rho	N	P	p.adj	qValueNoperm
LRIT1	0.11199011	346	5.105e−10	7.715e−06	1.166e−01
PDE6B	0.10626165	309	1.060e−08	7.449e−05	3.449e−01
TRPM6	0.08789351	366	1.479e−08	7.449e−05	3.449e−01
CPLX3	0.15258470	205	2.417e−08	9.130e−05	3.449e−01
CPB1	0.08431346	353	5.100e−08	1.541e−04	4.659e−01
DHDH	0.09543525	253	9.388e−07	6.987e−04	5.028e−01
TLE6	0.09539041	278	2.742e−07	4.363e−04	5.028e−01
GALNT5	0.07760340	354	1.659e−07	3.297e−04	5.028e−01
SPG11	0.06515057	377	7.444e−07	6.617e−04	5.028e−01
PHYH	0.07590208	317	9.709e−07	6.987e−04	5.028e−01
A2M	0.07001557	345	9.216e−07	6.987e−04	5.028e−01
TASOR2	0.06873580	361	6.544e−07	6.391e−04	5.028e−01
SMC4	0.07157356	350	5.795e−07	6.256e−04	5.028e−01
ABCC10	0.06670428	371	6.766e−07	6.391e−04	5.028e−01
AKNA	0.08200622	322	2.887e−07	4.363e−04	5.028e−01
AFF1	0.07165443	353	5.109e−07	5.939e−04	5.028e−01
THADA	0.06580886	369	8.604e−07	6.987e−04	5.028e−01
JAK3	0.09445217	290	1.745e−07	3.297e−04	5.028e−01
CFAP20DC	0.08094318	344	1.371e−07	3.297e−04	5.028e−01
TRPM1	0.07082851	362	4.268e−07	5.375e−04	5.028e−01
CRYAA2	0.14840183	177	3.211e−07	4.411e−04	5.028e−01
ADAMTS3	0.06708037	348	1.403e−06	8.833e−04	5.208e−01
SUSD5	0.06516177	342	2.431e−06	1.155e−03	5.208e−01
MUC20	0.08246624	276	1.915e−06	1.082e−03	5.208e−01
TDRD7	0.06456813	341	2.794e−06	1.206e−03	5.208e−01

Positive Rho Non−permulated



Negative Rho Non−permulated



Top Positive genes by P−value non−permulated

Gene	Rho	N	P	p.adj	qValueNoperm
LRIT1	0.11199011	346	5.105e−10	7.715e−06	1.166e−01
PDE6B	0.10626165	309	1.060e−08	7.449e−05	3.449e−01
TRPM6	0.08789351	366	1.479e−08	7.449e−05	3.449e−01
CPLX3	0.15258470	205	2.417e−08	9.130e−05	3.449e−01
CPB1	0.08431346	353	5.100e−08	1.541e−04	4.659e−01
GALNT5	0.07760340	354	1.659e−07	3.297e−04	5.028e−01
JAK3	0.09445217	290	1.745e−07	3.297e−04	5.028e−01
CFAP20DC	0.08094318	344	1.371e−07	3.297e−04	5.028e−01
TLE6	0.09539041	278	2.742e−07	4.363e−04	5.028e−01
AKNA	0.08200622	322	2.887e−07	4.363e−04	5.028e−01

Top Negative genes by P−value non−permulated

Gene	Rho	N	P	p.adj	qValueNoperm
NA	NA	NA	NA	NA	NA
NA.1	NA	NA	NA	NA	NA
NA.2	NA	NA	NA	NA	NA
NA.3	NA	NA	NA	NA	NA
NA.4	NA	NA	NA	NA	NA
NA.5	NA	NA	NA	NA	NA
NA.6	NA	NA	NA	NA	NA
NA.7	NA	NA	NA	NA	NA
NA.8	NA	NA	NA	NA	NA
NA.9	NA	NA	NA	NA	NA

GO_Biological_Process_2023 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
NADH Dehydrogenase Complex Assembly (GO:	0.16609623	48	6.950e−05	9.618e−02	NDUFB8:103 ECSIT:239 NDUFA9:398 TMEM126B:437 NDUFB6:488 TMEM126A:533
RNA Splicing, Via Transesterification Re	−0.11519753	106	4.318e−05	9.618e−02	RBMX2:3 PRPF40A:145 SNRNP27:179 SRRM1:186 XAB2:311 SRSF2:371
Mitochondrial Respiratory Chain Complex	0.16609623	48	6.950e−05	9.618e−02	NDUFB8:103 ECSIT:239 NDUFA9:398 TMEM126B:437 NDUFB6:488 TMEM126A:533
Monocarboxylic Acid Catabolic Process (G	0.25658984	20	7.142e−05	9.618e−02	PPARD:982 AGXT:1105 NUDT19:1116 LPIN1:1509 CYP26C1:1514 LPIN3:2539
Odontogenesis (GO:0042476)	−0.17410795	36	3.026e−04	3.260e−01	BCOR:377 MSX1:448 COL1A1:624 WNT10A:652 LAMB1:781 INHBA:854
Positive Regulation Of Amyloid−Beta Clea	−0.41474124	6	4.346e−04	3.902e−01	LRP1:258 IL4:324 TREM2:391 ROCK1:1359 APOE:2624 LRPAP1:2699
Alpha−Amino Acid Catabolic Process (GO:1	0.18166929	29	7.127e−04	4.799e−01	HIBCH:471 HMGCL:612 ALDH4A1:676 TAT:1748 KMO:1858 GOT2:1948
mRNA Processing (GO:0006397)	−0.08562310	132	7.030e−04	4.799e−01	RBMX2:3 PRPF40A:145 SNRNP27:179 SRRM1:186 XAB2:311 SRSF2:371
Visual Perception (GO:0007601)	0.10239206	90	8.027e−04	4.804e−01	TRPM1:12 CRYBB2:27 CRYAA:43 CRYBB1:139 CRYZ:148 VAX2:154
mRNA Splicing, Via Spliceosome (GO:00003	−0.07884419	135	1.605e−03	5.086e−01	RBMX2:3 PRPF40A:145 SNRNP27:179 SRRM1:186 XAB2:311 SRSF2:371
Mitochondrial Respiratory Chain Complex	0.10460349	77	1.529e−03	5.086e−01	NDUFB8:103 ECSIT:239 NDUFA9:398 TMEM126B:437 NDUFB6:488 TMEM126A:533
Negative Regulation Of Intrinsic Apoptot	−0.25495752	14	9.582e−04	5.086e−01	KDM1A:542 TAF9:636 ARMC10:703 TAF9B:820 PTTG1IP:1004 ING2:1175
Negative Regulation Of Intrinsic Apoptot	−0.28883582	10	1.564e−03	5.086e−01	KDM1A:542 TAF9:636 TAF9B:820 ING2:1175 ZNF385A:2967 SIRT1:3526
Negative Regulation Of Microtubule Polym	−0.18116135	27	1.126e−03	5.086e−01	TPX2:267 TRIM54:323 MAP2:446 MAPRE1:449 KATNB1:465 NAV3:474
Negative Regulation Of Protein Catabolic	−0.13616862	45	1.588e−03	5.086e−01	AZIN2:295 HMGCR:533 HSP90AB1:578 PTPN3:600 AGAP2:622 SERPINB12:908
Regulation Of Intrinsic Apoptotic Signal	−0.28883582	10	1.564e−03	5.086e−01	KDM1A:542 TAF9:636 TAF9B:820 ING2:1175 ZNF385A:2967 SIRT1:3526
Sister Chromatid Segregation (GO:0000819	0.16818738	30	1.438e−03	5.086e−01	SMC4:14 KIF18B:115 SPAG5:253 ZWINT:272 KIFC1:333 ESPL1:1008
Connective Tissue Development (GO:006144	−0.14893398	37	1.729e−03	5.115e−01	HYAL2:78 VPS13B:434 SOX6:640 CHI3L1:648 SPTLC2:765 SNORC:776
Mitochondrial Electron Transport, NADH T	0.16202346	31	1.804e−03	5.115e−01	NDUFB8:103 NDUFA9:398 NDUFV3:403 NDUFB6:488 NDUFA6:700 NDUFB5:702
Positive Regulation Of Type I Interferon	−0.12094645	55	1.937e−03	5.217e−01	FLOT1:259 DHX58:345 POLR3B:382 TLR7:452 RAB2B:476 TRIM56:508
Regulation Of Amyloid−Beta Clearance (GO	−0.22875995	15	2.162e−03	5.546e−01	LRP1:258 IL4:324 TREM2:391 HMGCR:533 ROCK1:1359 IFNG:1538
7−Methylguanosine Cap Hypermethylation (	−0.35037507	4	1.523e−02	6.121e−01	SNRPG:1287 SNRPD3:1452 SNRPD1:1583 TGS1:4673 NA NA
G Protein−Coupled Glutamate Receptor Sig	0.22107720	11	1.113e−02	6.121e−01	TRPM1:12 GRM6:271 HOMER2:550 GRM8:630 GRM2:672 GRM7:1174
Golgi To Plasma Membrane Transport (GO:0	0.10979990	41	1.505e−02	6.121e−01	GOLGA4:40 MACF1:863 OSBPL5:1519 ANK3:1653 PREPL:1667 NSF:1809
RNA Capping (GO:0036260)	−0.35037507	4	1.523e−02	6.121e−01	SNRPG:1287 SNRPD3:1452 SNRPD1:1583 TGS1:4673 NA NA
RNA Processing (GO:0006396)	−0.07231650	122	5.915e−03	6.121e−01	SRRM1:186 DIS3:253 PPP1R9B:354 SRSF2:371 CELF2:503 METTL3:632
RNA−mediated Gene Silencing (GO:0031047)	0.17432845	22	4.659e−03	6.121e−01	MOV10L1:80 PIWIL1:130 FKBP6:213 TNRC6B:470 DICER1:1003 MOV10:1534
UV Protection (GO:0009650)	−0.30524021	7	5.164e−03	6.121e−01	XPA:436 SCARA3:1591 FEN1:2007 MFAP4:3551 SDF4:3667 MC1R:4515
Aerobic Respiration (GO:0009060)	0.11215933	53	4.772e−03	6.121e−01	NDUFB8:103 NDUFA9:398 NDUFV3:403 NDUFB6:488 NDUFA6:700 NDUFB5:702
Ammonium Homeostasis (GO:0097272)	0.32057971	5	1.304e−02	6.121e−01	RHAG:308 RHCG:429 RHD:905 RHCE:1757 RHBG:10080 NA
Ammonium Transmembrane Transport (GO:007	0.21812367	10	1.693e−02	6.121e−01	RHAG:308 RHCG:429 RHD:905 RHCE:1757 SLC12A4:3372 SLC12A2:4964
Antigen Receptor−Mediated Signaling Path	−0.07441540	97	1.146e−02	6.121e−01	NCKAP1L:93 NFKBID:239 NCOR2:269 CD3G:358 RBCK1:390 THY1:413
Astrocyte Activation (GO:0048143)	−0.24132759	12	3.800e−03	6.121e−01	C5AR1:241 LRP1:258 TREM2:391 IFNG:1538 TNF:1567 PSEN1:1778
Axonemal Central Apparatus Assembly (GO:	−0.36772379	5	4.404e−03	6.121e−01	DNAJB13:79.5 RSPH9:1056 HYDIN:1109 SPEF1:1833 SPAG17:5903 NA
Cardiac Muscle Cell Myoblast Differentia	0.33181586	5	1.018e−02	6.121e−01	RBPJ:300 TBXT:654 NRG1:1444 SRF:5118 NOTCH1:5158 NA
Cardiac Muscle Hypertrophy (GO:0003300)	0.29256544	6	1.307e−02	6.121e−01	CSRP3:968 HDAC4:1247 RYR2:1406 TCAP:2584 HTR2B:3103 OCRL:9399
Cartilage Development (GO:0051216)	−0.10455077	47	1.322e−02	6.121e−01	HYAL2:78 SOX6:640 CHI3L1:648 SNORC:776 SCX:780 COL11A2:808
Cellular Lipid Catabolic Process (GO:004	0.16295438	23	6.842e−03	6.121e−01	PLD1:123 ANGPTL3:127 PPARD:982 NUDT19:1116 LPIN1:1509 SMPDL3B:2200
Cellular Response To Amino Acid Starvati	−0.12774648	37	7.201e−03	6.121e−01	CDKN1A:332 DEPDC5:717 RRAGD:832 MTOR:1026 WDR24:1128 DAPL1:2086
Cellular Response To Bacterial Lipopepti	0.33241262	5	1.005e−02	6.121e−01	TLR10:529 TLR1:1248 CD36:2773 TLR2:3942 TLR6:4096 NA

EnrichmentHsSymbolsFile2 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
IBRAHIM_NRF1_UP	−0.06673889	357	1.597e−05	5.614e−02	NUP62:12 MAP4:25 EDEM3:63 DNAJC10:65 MRGBP:102 KIF3B:133
WP_MITOCHONDRIAL_COMPLEX_I_ASSEMBLY_MODE	0.18123310	47	1.731e−05	5.614e−02	NDUFB8:103 ECSIT:239 NDUFV3:403 TMEM126B:437 NDUFB6:488 NDUFA6:700
REACTOME_COMPLEX_I_BIOGENESIS	0.16484010	48	7.829e−05	1.270e−01	NDUFB8:103 ECSIT:239 NDUFA9:398 NDUFV3:403 TMEM126B:437 NDUFB6:488
WP_INTERFERON_TYPE_I_SIGNALING_PATHWAYS	−0.17646211	42	7.628e−05	1.270e−01	JAK1:198 MAP2K3:592 IRF9:620 EIF4E:749 GAB2:762 EIF4EBP1:968
KIM_TIAL1_TARGETS	−0.22889504	24	1.039e−04	1.349e−01	SMARCE1:87 PLSCR3:250 ADSL:343 BDH1:889 ADGRG1:983 FGL1:1020
IBRAHIM_NRF2_UP	−0.04836247	450	4.667e−04	1.791e−01	NCOA7:24 STK40:33 EDEM3:63 DNAJC10:65 PPT1:128 KIF3B:133
NIKOLSKY_BREAST_CANCER_12Q24_AMPLICON	0.27650944	15	2.091e−04	1.791e−01	EP400:87 P2RX2:349 SFSWAP:681 POLE:1101 GOLGA3:1239 GALNT9:1515
CHEN_LVAD_SUPPORT_OF_FAILING_HEART_UP	−0.10679159	87	5.812e−04	1.791e−01	HYAL2:78 ACTA1:91 SLC38A2:240 FOSL2:279 CDKN1A:332 LAMB1:781
BONCI_TARGETS_OF_MIR15A_AND_MIR16_1	−0.11283135	78	5.759e−04	1.791e−01	PCDHAC2:7 HAS2:363 CDCA4:565 PTPN3:600 BTRC:634 KIF21A:912
LU_EZH2_TARGETS_UP	−0.06442789	231	7.645e−04	1.791e−01	SCAND1:50 HYAL2:78 OASL:110 HDGFL2:231 FAM25A:245 FLOT1:259
FISCHER_DIRECT_P53_TARGETS_META_ANALYSIS	−0.06085034	256	8.283e−04	1.791e−01	RFK:21 RRAD:104 CNOT6:118 TANC1:210 LRP1:258 APAF1:320
FLORIO_NEOCORTEX_BASAL_RADIAL_GLIA_DN	0.07801637	172	4.258e−04	1.791e−01	SMC4:14 DLGAP5:74 KIF18B:115 SPAG5:253 ZWINT:272 MIS18BP1:315
REACTOME_ADAPTIVE_IMMUNE_SYSTEM	−0.04225577	581	5.442e−04	1.791e−01	TUBAL3:79.5 SNAP23:81 PSMD8:131 KIF3B:133 ASB13:192 HERC3:206
REACTOME_HOST_INTERACTIONS_OF_HIV_FACTOR	−0.10056151	93	8.119e−04	1.791e−01	NUP62:12 PSMD8:131 PSMB10:453 BTRC:634 ATP6V1H:668 PSMD3:822
REACTOME_INNATE_IMMUNE_SYSTEM	−0.03593705	770	7.711e−04	1.791e−01	GLIPR1:27 NHLRC3:36 SLC2A3:37 AHSG:43 ASAH1:72 SNAP23:81
REACTOME_METABOLISM_OF_POLYAMINES	−0.16072400	42	3.146e−04	1.791e−01	PSMD8:131 AZIN2:295 PSMB10:453 PSMD3:822 PSMB11:951 ODC1:1221
REACTOME_REGULATION_OF_MRNA_STABILITY_BY	−0.13255877	61	3.450e−04	1.791e−01	PSMD8:131 SET:252 DIS3:253 PSMB10:453 PSMD3:822 EXOSC7:869
REACTOME_TP53_REGULATES_TRANSCRIPTION_OF	−0.15491842	40	7.005e−04	1.791e−01	CNOT6:118 CNOT11:170 CDKN1A:332 PLK2:361 CNOT3:599 RBL1:744
REACTOME_THE_ROLE_OF_GTSE1_IN_G2_M_PROGR	−0.14241656	54	2.962e−04	1.791e−01	TUBAL3:79.5 PSMD8:131 CDKN1A:332 MAPRE1:449 PSMB10:453 HSP90AB1:578
REACTOME_CELLULAR_RESPONSE_TO_CHEMICAL_S	−0.08475338	151	3.318e−04	1.791e−01	PRDX2:20 PSMD8:131 NCOR2:269 SCO2:305 CDKN1A:332 PSMB10:453
WP_TYPE_II_INTERFERON_SIGNALING	−0.19085371	30	2.976e−04	1.791e−01	JAK1:198 IRF9:620 CIITA:899 IFIT2:949 H4C14:955 CYBB:956
FULCHER_INFLAMMATORY_RESPONSE_LECTIN_VS_	−0.04629433	493	4.642e−04	1.791e−01	GLIPR1:27 KCNK6:75 SUN2:86 CNPY2:97 TAPT1:130 ACSL3:152
DIAZ_CHRONIC_MYELOGENOUS_LEUKEMIA_UP	−0.03057504	1149	5.660e−04	1.791e−01	GMCL1:49 DLD:59 DRAP1:64 SMARCA5:66 SNAP23:81 CD47:82
UDAYAKUMAR_MED1_TARGETS_DN	−0.07365359	202	3.167e−04	1.791e−01	CDKN1C:23 SLC2A3:37 ELF3:85 GABARAPL1:159 HOXC13:200 UPF1:205
SATO_SILENCED_BY_METHYLATION_IN_PANCREAT	−0.05528054	326	6.284e−04	1.791e−01	CDKN1C:23 CCL5:123 HOXC13:200 HERC3:206 KLK11:214 NCOR2:269
NAKAYAMA_SOFT_TISSUE_TUMORS_PCA1_UP	−0.12322436	62	7.967e−04	1.791e−01	FAP:69 NCKAP1L:93 CCL5:123 LUM:375 HLA−DPA1:694 HLA−DRA:868
BORCZUK_MALIGNANT_MESOTHELIOMA_UP	−0.06324231	239	7.817e−04	1.791e−01	PPT1:128 PSMD8:131 RPS15A:167 TPX2:267 UBQLN2:294 COPA:352
SMIRNOV_RESPONSE_TO_IR_6HR_UP	−0.08850775	135	3.913e−04	1.791e−01	RRAD:104 ANO3:190 APAF1:320 CDKN1A:332 PLK2:361 TOB1:420
BIOCARTA_PLATELETAPP_PATHWAY	−0.26034396	14	7.443e−04	1.791e−01	F9:509 F2:512 PLAU:619 F11:723 PLAT:734 COL4A4:770
KEGG_UBIQUITIN_MEDIATED_PROTEOLYSIS	−0.09461020	106	7.737e−04	1.791e−01	CBLC:124 HERC3:206 SYVN1:325 UBE2J1:381 FBXO4:468 BTRC:634
GRAESSMANN_RESPONSE_TO_MC_AND_SERUM_DEPR	−0.07249715	174	9.912e−04	1.814e−01	CPEB3:45 ZNF503:67 LGALS3BP:77 ACTB:146 SEPTIN6:207 NMNAT1:225
REACTOME_PD_1_SIGNALING	−0.25413225	14	9.939e−04	1.814e−01	CD3G:358 CD274:519.5 HLA−DPA1:694 CD3E:823 HLA−DRA:868 LCK:1021
REACTOME_HOMOLOGY_DIRECTED_REPAIR	0.09250409	107	9.581e−04	1.814e−01	RNF168:191 CLSPN:230 HERC2:280 BRCA1:395 WRN:418 POLH:454
REACTOME_CELLULAR_RESPONSES_TO_STIMULI	−0.03992379	589	1.007e−03	1.814e−01	RPS26:3 NUP62:12 PRDX2:20 TUBAL3:79.5 CA9:109 PSMD8:131
REACTOME_CELLULAR_RESPONSE_TO_STARVATION	−0.10211568	89	8.780e−04	1.814e−01	RPS26:3 RPL7:157 RPS15A:167 ATP6V1H:668 DEPDC5:717 TCIRG1:779
BECKER_TAMOXIFEN_RESISTANCE_UP	−0.15433338	38	9.973e−04	1.814e−01	LGALS3BP:77 ACTA1:91 EBAG9:191 IRF9:620 BTG3:1254 TAGLN:1258
HERNANDEZ_MITOTIC_ARREST_BY_DOCETAXEL_2_	−0.13379826	46	1.698e−03	2.173e−01	CD47:82 HHEX:155 LUM:375 PLAU:619 PLAT:734 KLF9:835
NIKOLSKY_BREAST_CANCER_8P12_P11_AMPLICON	0.13167007	50	1.283e−03	2.173e−01	NSD3:76 IDO2:77 RAB11FIP1:99 ADAM9:144 DKK4:211 TACC1:226
REACTOME_CYTOKINE_SIGNALING_IN_IMMUNE_SY	−0.03989573	555	1.404e−03	2.173e−01	NUP62:12 OASL:110 CCL5:123 PSMD8:131 TNFRSF9:166 JAK1:198
REACTOME_SIGNALING_BY_ROBO_RECEPTORS	−0.07918372	132	1.706e−03	2.173e−01	RPS26:3 PSMD8:131 RPL7:157 RPS15A:167 PSMB10:453 CAP2:514

DisGeNET Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Cataract, Pulverulent	0.41370118	9	1.727e−05	7.021e−02	CRYBB2:27 CRYBB1:139 HSF4:150 GJA8:248 BFSP2:656 VIM:1134
Glioma	−0.03203929	1646	2.864e−05	7.021e−02	TYMP:11 NUP62:12 PDGFB:14 PRDX2:20 CDKN1C:23 GLIPR1:27
Nuclear cataract	0.21571935	32	2.428e−05	7.021e−02	CRYBB2:27 CRYAA:43 CRYBB1:139 HSF4:150 CRYBB3:227 GJA8:248
Nuclear non−senile cataract	0.23396434	29	1.306e−05	7.021e−02	CRYBB2:27 CRYAA:43 CRYBB1:139 CRYBB3:227 GJA8:248 MIP:415
CATARACT, AUTOSOMAL DOMINANT	0.37431299	10	4.158e−05	8.154e−02	CRYAA2:11 CRYBB2:27 CRYAA:43 CRYBB1:139 GJA8:248 MIP:415
Medullary carcinoma of thyroid	−0.08957154	173	5.087e−05	8.313e−02	TYMP:11 MET:181 TNFSF10:339 ALCAM:356 PROM1:359 RXFP2:457
Cataract, Central Saccular, With Sutural	0.46084046	6	9.259e−05	8.911e−02	CRYBB2:27 GJA8:248 MIP:415 BFSP2:656 CRYBA1:662 CRYGS:1486
Malaria, Falciparum	−0.13584876	68	1.091e−04	8.911e−02	MST1:62 CCL5:123 IL4:324 CHI3L1:648 LMLN:666 CD27:690
Carcinoma of lung	−0.02908051	1859	6.487e−05	8.911e−02	CFTR:3 FKBP10:9 TYMP:11 PDGFB:14 DTYMK:16 NABP2:19
Defective tooth enamel	0.18722671	36	1.021e−04	8.911e−02	GTF2IRD1:287 CENPJ:382 PEX6:635 PLK4:866 PEX1:1000 PLEC:1084
Dystrophic tooth enamel	0.18722671	36	1.021e−04	8.911e−02	GTF2IRD1:287 CENPJ:382 PEX6:635 PLK4:866 PEX1:1000 PLEC:1084
Primary malignant neoplasm of lung	−0.02991116	1703	7.425e−05	8.911e−02	CFTR:3 FKBP10:9 TYMP:11 DTYMK:16 NABP2:19 PRDX2:20
Abnormality of dental enamel	0.17120474	40	1.807e−04	1.266e−01	GTF2IRD1:287 CENPJ:382 PEX6:635 PLK4:866 PEX1:1000 PLEC:1084
Malignant neoplasm of lung	−0.02756778	1828	1.695e−04	1.266e−01	CFTR:3 FKBP10:9 TYMP:11 DTYMK:16 NABP2:19 PRDX2:20
Non−Small Cell Lung Carcinoma	−0.02821184	1606	2.662e−04	1.740e−01	CFTR:3 TYMP:11 NUP62:12 PRDX2:20 CDKN1C:23 SLC2A3:37
CATARACT, COPPOCK−LIKE	0.34576715	9	3.284e−04	1.789e−01	CRYBB2:27 CRYBB1:139 GJA8:248 BFSP2:656 VIM:1134 GJA3:2824
Neoplasm Metastasis	−0.02218974	2915	3.020e−04	1.789e−01	CFTR:3 C1GALT1:8 TYMP:11 NUP62:12 PDGFB:14 PRDX2:20
Sacral dimples	0.22156494	22	3.224e−04	1.789e−01	GTF2IRD1:287 NSD2:455 NELFA:556 CLIP2:1187 LETM1:1192 ASXL1:1331
Liver carcinoma	−0.02268547	2600	3.873e−04	1.999e−01	CFTR:3 C1GALT1:8 TYMP:11 NUP62:12 PDGFB:14 NABP2:19
Coxiella burnetii Infection	−0.40729521	6	5.503e−04	2.627e−01	FLOT1:259 HMGCR:533 LDLR:537 LSS:1189 TNF:1567 DHCR24:4223
First myocardial infarction	−0.31499377	10	5.626e−04	2.627e−01	F2:512 LDLR:537 CPB2:569 PLAT:734 PON2:797 IL1RN:1342
Stomach Diseases	−0.21389792	21	6.928e−04	3.088e−01	S100A8:415 HGF:490 GCG:914 AQP3:979 ODC1:1221 SMOX:1489
Dermatitis, Irritant	−0.23286065	17	8.892e−04	3.459e−01	CXCR3:229 IL4:324 CD80:418 HLA−DPA1:694 TRIM10:1286 TNF:1567
Idiopathic pulmonary arterial hypertensi	−0.06472618	226	8.413e−04	3.459e−01	CD47:82 CA9:109 CCL5:123 POU5F1:136 PAM16:213 APLNR:330
Malignant tumor of colon	−0.02609539	1542	9.171e−04	3.459e−01	CFTR:3 C1GALT1:8 TYMP:11 DTYMK:16 PRDX2:20 CDKN1C:23
Nausea	−0.14619005	43	9.173e−04	3.459e−01	TYMP:11 UGT1A1:608 NT5C:752 DHDDS:886 HTR3A:1249 RET:1283
Anemia, hereditary spherocytic hemolytic	0.41384232	5	1.352e−03	3.850e−01	EPB42:332 ANK1:468 SPTA1:756 SLC4A1:2195 SPTB:2665 NA
Night blindness, congenital stationary	0.19207837	23	1.433e−03	3.850e−01	PDE6B:2 TRPM1:12 USP11:113 GRM6:271 CACNA2D4:360 ABCA4:641
Cataract microcornea syndrome	0.30463778	9	1.553e−03	3.850e−01	CRYBB2:27 CRYAA:43 CRYBB1:139 GJA8:248 CRYBA4:1140 CRYGD:3210
Congenital total cataract	0.24276855	15	1.134e−03	3.850e−01	CRYBB2:27 CRYAA:43 HSF4:150 GJA8:248 GCNT2:371 MIP:415
HIV Infections	−0.03908041	591	1.331e−03	3.850e−01	PDGFB:14 DTYMK:16 HNF1A:35 FAP:69 CD47:82 CCL5:123
Lip and Oral Cavity Carcinoma	−0.04832152	372	1.482e−03	3.850e−01	CDKN1C:23 LGALS3BP:77 CA9:109 CCL5:123 CTHRC1:315 IL4:324
Neuroblastoma	−0.02670575	1330	1.462e−03	3.850e−01	TYMP:11 NUP62:12 DTYMK:16 PRDX2:20 CDKN1C:23 NCOA7:24
Osteosarcoma of bone	−0.03609205	721	1.134e−03	3.850e−01	PDGFB:14 DTYMK:16 PRDX2:20 CD47:82 CA9:109 CDKN3:117
Pelvic kidney	0.28815276	10	1.604e−03	3.850e−01	GTF2IRD1:287 CLIP2:1187 FGFR1:1540 RFC2:1551 GTF2I:1666 STRA6:2945
Pierre Robin Syndrome	−0.19795562	22	1.312e−03	3.850e−01	DGCR2:162 COL11A2:808 CD6:919 DGCR6:1202 AMER1:1220 MYMK:1265
Prostate carcinoma	−0.02115122	2344	1.454e−03	3.850e−01	CFTR:3 NUP62:12 PDGFB:14 DTYMK:16 PRDX2:20 CYP7B1:22
Retinal pigment epithelial atrophy	0.24704877	14	1.374e−03	3.850e−01	CERKL:334 PRPH2:606 ABCA4:641 CNGB3:836 ELOVL4:2098 RS1:2371
Secondary malignant neoplasm of lymph no	−0.03105033	932	1.610e−03	3.850e−01	TYMP:11 PRDX2:20 CDKN1C:23 GATA5:42 DLD:59 MST1:62
Spherocytosis	0.41384232	5	1.352e−03	3.850e−01	EPB42:332 ANK1:468 SPTA1:756 SLC4A1:2195 SPTB:2665 NA

MGI_Mammalian_Phenotype_Level_4 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
MP0002169 no abnormal phenotype	−0.02925535	1083	3.134e−03	4.825e−01	CFTR:3 PCDHAC2:7 FAP:69 SUN2:86 RABGAP1L:98 CBLC:124
MP0005551 abnormal eye electrophysiolog	0.07914660	136	1.613e−03	4.825e−01	PDE6B:2 CPLX3:4 TRPM1:12 GABRR1:50 CDHR1:55 NAGLU:96
MP0003252 abnormal bile duct	0.18139323	21	4.063e−03	4.825e−01	PKHD1:412 ABCB11:651 ABCB4:879 PKD2:1518 NR1H4:1591 ABCG5:1681
MP0002873 normal phenotype	−0.02925240	1088	3.087e−03	4.825e−01	CFTR:3 PCDHAC2:7 FAP:69 SUN2:86 RABGAP1L:98 CBLC:124
MP0002452 abnormal antigen presenting	−0.03227453	539	1.419e−02	6.126e−01	CFTR:3 RFK:21 GPR183:89 NCKAP1L:93 SASH3:100 TNFRSF9:166
MP0001286 abnormal eye development	−0.06344121	140	1.036e−02	6.126e−01	CDKN1C:23 MAB21L2:70 VAX1:141 HHEX:155 PITX3:178 DISP1:255
MP0001853 heart inflammation	−0.14565495	25	1.186e−02	6.126e−01	NCKAP1L:93 RELA:693 GJA1:985 IRAK1:1144 TREX1:1763 CHD2:2122
MP0001819 abnormal immune cell	−0.02750516	822	1.247e−02	6.126e−01	CFTR:3 RFK:21 CD47:82 GPR183:89 NCKAP1L:93 SASH3:100
MP0003221 abnormal cardiomyocyte apopto	−0.12869091	33	1.070e−02	6.126e−01	TRIM54:323 DNAJA3:644 BACH1:794 NOX4:1379 SLC8A1:1515 MAPK9:1977
MP0001666 abnormal nutrient absorption	0.11299319	40	1.368e−02	6.126e−01	DRD2:105 SOAT2:231 ABCG8:736 ABCB4:879 SLC23A1:1289 ABCG5:1681
MP0002420 abnormal adaptive immunity	−0.02737995	818	1.306e−02	6.126e−01	CFTR:3 RFK:21 CD47:82 GPR183:89 NCKAP1L:93 SASH3:100
MP0005085 abnormal gallbladder physiolo	0.14276802	24	1.566e−02	6.197e−01	GPBAR1:277 ABCB11:651 ABCG8:736 ABCB4:879 CYP27A1:959 F2RL1:1285
MP0002723 abnormal immune serum	−0.03009096	595	1.697e−02	6.200e−01	LGALS3BP:77 GPR183:89 NCKAP1L:93 SASH3:100 CCL5:123 SCARB1:132
MP0000001 mammalian phenotype	−0.01819359	2791	2.628e−02	6.912e−01	CFTR:3 PCDHAC2:7 TYMP:11 PDGFB:14 CDKN1C:23 EXT2:26
MP0001929 abnormal gametogenesis	0.03251584	410	2.884e−02	6.912e−01	PHYH:21 PHC2:66 MOV10L1:80 RLIM:128 PIWIL1:130 FOXN1:145
MP0002932 abnormal joint morphology	−0.06601638	98	2.499e−02	6.912e−01	HYAL2:78 SYVN1:325 HAS2:363 COL1A1:624 LRP6:698 PLAT:734
MP0002272 abnormal nervous system	0.05049697	175	2.301e−02	6.912e−01	PHYH:21 GABRR1:50 DRD2:105 KCNC2:254 KCNQ3:323 KCNQ2:408
MP0000432 abnormal head morphology	−0.03971238	285	2.410e−02	6.912e−01	CFTR:3 WNT9B:76 HYAL2:78 VAX1:141 SP3:184 FGF6:242
MP0000653 abnormal sex gland	0.03210069	457	2.342e−02	6.912e−01	MOV10L1:80 DRD2:105 PIWIL1:130 FOXN1:145 FANCM:192 FKBP6:213
MP0010094 abnormal chromosome stability	0.08391860	57	2.910e−02	6.912e−01	FOXN1:145 FANCG:224 BRCA1:395 ASZ1:728 BARD1:838 BUB1B:953
MP0008438 abnormal cutaneous collagen	0.16376341	14	3.405e−02	7.033e−01	COL3A1:167 COL5A2:1251 PPIB:1934 FBN1:3222 LOX:3465 COL5A1:3793
MP0003436 decreased susceptibility to	−0.09968916	38	3.396e−02	7.033e−01	ALOX5AP:139 CLEC5A:221 SYVN1:325 CDKN1A:332 PLAU:619 DUSP2:786
MP0002118 abnormal lipid homeostasis	−0.03291086	380	3.263e−02	7.033e−01	CFTR:3 CYP7B1:22 HNF1A:35 AHSG:43 GPD2:57 ASAH1:72
MP0002080 prenatal lethality	−0.01985391	1206	3.730e−02	7.269e−01	C1GALT1:8 PDGFB:14 RFK:21 CDKN1C:23 EXT2:26 HNF1A:35
MP0005379 endocrine/exocrine gland phen	−0.14533493	17	3.826e−02	7.269e−01	CFTR:3 CDKN1C:23 PRLR:1184 MC5R:2215 PAX7:2327 E2F1:2615
MP0002128 abnormal blood circulation	−0.03147324	378	4.151e−02	7.584e−01	C1GALT1:8 PDGFB:14 MET:181 LRP1:258 TRIM54:323 CDKN1A:332
MP0001697 abnormal embryo size	−0.03145925	364	4.524e−02	7.960e−01	DLD:59 KIF3B:133 MET:181 HS3ST1:224 DISP1:255 SMAD1:349
MP0002210 abnormal sex determination	0.03328756	302	5.206e−02	8.153e−01	MOV10L1:80 DRD2:105 PIWIL1:130 FOXN1:145 FANCM:192 FKBP6:213
MP0002138 abnormal hepatobiliary system	0.19592040	8	5.512e−02	8.153e−01	PKHD1:412 ABCB4:879 SOCS2:1746 ARG1:2060 MAP3K14:4724 ACVRL1:6713
MP0005253 abnormal eye physiology	0.05845015	89	5.836e−02	8.153e−01	TRPM1:12 NR2E3:247 GJA8:248 EGFLAM:264 MIP:415 PRPH2:606
MP0008875 abnormal xenobiotic pharmacok	0.13886536	16	5.474e−02	8.153e−01	CYP1A1:484 CYP1A2:530 MTF1:1052 AS3MT:2214 AHR:2987 COMT:3602
MP0002106 abnormal muscle physiology	−0.04767613	147	4.852e−02	8.153e−01	ACTA1:91 CACNA2D1:212 FGF6:242 CDKN1A:332 RICTOR:546 FBXO32:594
MP0002653 abnormal ependyma morphology	−0.18307679	9	5.734e−02	8.153e−01	VAX1:141 E2F5:913 DLG5:1086 HYDIN:1109 NUMB:2753 DLL3:2838
MP0006054 spinal hemorrhage	−0.19917910	8	5.120e−02	8.153e−01	C1GALT1:8 COL1A1:624 PSEN1:1778 C1GALT1C1:3804 SPHK1:4007 FLI1:5333
MP0000609 abnormal liver physiology	−0.02926623	360	6.384e−02	8.193e−01	HNF1A:35 GPD2:57 HYAL2:78 NCKAP1L:93 SCARB1:132 PITX3:178
MP0000604 amyloidosis	−0.08962424	33	7.549e−02	8.193e−01	NCKAP1L:93 LDLR:537 MSR1:986 DLST:1599 PSEN1:1778 MAPT:2090
MP0001765 abnormal ion homeostasis	0.05263621	99	7.248e−02	8.193e−01	TRPM6:3 ATP2A2:229 CLDN16:426 RHCG:429 NEDD4L:505 PTH1R:562
MP0001845 abnormal inflammatory respons	−0.02480243	475	7.478e−02	8.193e−01	CFTR:3 C1GALT1:8 LGALS3BP:77 NCKAP1L:93 SASH3:100 ALOX5AP:139
MP0008877 abnormal DNA methylation	0.13715026	14	7.589e−02	8.193e−01	MOV10L1:80 UHRF1:275 ASZ1:728 ARID4A:995 RNF4:3758 MAEL:4067
MP0004808 abnormal hematopoietic stem	0.06936060	58	6.890e−02	8.193e−01	EP400:87 MPL:186 CSF3R:442 ITGA2B:561 ARID4A:995 MECOM:1121

tissue_specific Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
eye.development	−0.104774103	57	6.361e−03	3.435e−01	ELF3:85 VAX1:141 PITX3:178 SMAD1:349 LUM:375 MSX1:448
amygdala	−0.143304291	8	1.606e−01	8.187e−01	PCDH15:56 SLC38A8:2596 NEUROD2:2846 PTPRZ1:4325 TNR:5009 BCAN:5833
atrium.auricle	0.026920169	89	3.821e−01	8.187e−01	TNNI3K:46 XIRP1:826 LMOD3:878 CSRP3:968 BMP10:983 MYOM2:1012
blood	0.025186711	266	1.630e−01	8.187e−01	JAK3:8 BPI:194 RHAG:308 TREML2:329 EPB42:332 DHX34:343
breast	−0.095224137	14	2.176e−01	8.187e−01	GSC:2183 CIDEC:3278.5 ADIPOQ:3671 LEP:4152 FABP4:4389 ABCB5:4722
C1.spinal.cord	−0.021171415	171	3.437e−01	8.187e−01	CTNNA3:203 ARHGAP22:293 PLD4:340 TREM2:391 TMEM235:696 ACSBG1:721
cerebellar.hemisphere	0.017615701	459	2.067e−01	8.187e−01	PHTF1:78 ADCY1:137 VAX2:154 NBEA:163 JMJD7:172 CADM3:190
cortex.kidney	0.034900064	110	2.085e−01	8.187e−01	SLC5A12:36 CLCNKB:116 CUBN:367 PKHD1:412 CLDN16:426 DCDC2:560
esophagogastric.junction	−0.090057109	8	3.779e−01	8.187e−01	HOXA4:853 ADCY5:3019 NKX6−1:3649 CHRM3:3925 BARX1:4380 COL4A5:6510
esophagus.mucosa	0.028850843	213	1.511e−01	8.187e−01	ADGRF1:23 SOX15:89 RAB11FIP1:99 DSG3:102 FOXN1:145 CRB3:170
esophagus.muscularis.mucosa	−0.095586325	26	9.198e−02	8.187e−01	EPHA7:257 FAM83D:268 PALLD:739 S1PR3:1029 KCNMB1:1967 SYNPO2:2064
fallopian.tube	0.035630554	48	3.942e−01	8.187e−01	ADAMTS3:24 CCDC17:58 ADGRG4:244 GMNC:296 GREB1L:1492 GASK1A:1972
greater.omentum	−0.054279941	32	2.886e−01	8.187e−01	PRRT4:10 GFPT2:187 MEST:378 TCF15:1074 FAM89A:1593 RBP7:2780
hippocampus.proper	−0.100501204	9	2.966e−01	8.187e−01	NTS:2270 GRP:2551 NEUROD2:2846 NEUROG3:4107 FEZF2:5452 OLIG2:5905
hypothalamus	−0.057042286	58	1.340e−01	8.187e−01	PCDH15:56 GSX1:834 SYNGR3:1006 NGB:1017 HAP1:1079 PMCH:1262
liver	0.025383849	322	1.235e−01	8.187e−01	ACY1:26 THPO:33 SERPINA7:48 ACOX2:52 ADGRG7:56 IDO2:77
ovary	0.027500598	98	3.492e−01	8.187e−01	CPZ:216 PLIN5:477 TSHZ3:565 CCDC68:589 NLGN4X:626 SYTL4:675
pancreas	0.043123956	93	1.525e−01	8.187e−01	CPB1:5 TMED6:125 ERP27:159 CELA3B:208 KCNK16:380 CTRC:414
pituitary.gland	0.017640475	202	3.923e−01	8.187e−01	OTOG:68 IGSF1:86 VEPH1:101 DRD2:105 REC8:386 RIMBP2:404
prostate	0.046287758	37	3.307e−01	8.187e−01	NR2E3:247 P2RX2:349 SCUBE2:460 CPAMD8:520 SHROOM1:613 CREB3L4:1529
putamen	−0.088199577	14	2.535e−01	8.187e−01	ANO3:190 SLC18A3:1457 GRM3:1776 OPALIN:2884 FAM237A:3160 GBX2:4190
testis	−0.012644560	1339	1.507e−01	8.187e−01	DNAH2:3 C11orf42:47 BTG4:53 DIAPH3:58 DNAI2:74 DNAJB13:79.5
thyroid	0.029958411	147	2.133e−01	8.187e−01	TLE6:9 CLCNKA:57 PRTG:61 IDO2:77 DGKI:153 CRYBB3:227
transverse.colon	−0.022846735	144	3.474e−01	8.187e−01	HNF1A:35 TUBAL3:79.5 ATP10B:122 R3HDML:140 PYY:447 LEFTY1:496
uterus	0.043245483	53	2.773e−01	8.187e−01	CLEC4F:263 TEX15:352 ADAMTS19:731 CHRDL2:899 HOXD4:1269 MRGPRF:1503
vagina	0.043551814	89	1.574e−01	8.187e−01	DSG3:102 FOXN1:145 PGLYRP4:250 SLC18A2:310 RHCG:429 DSC2:602
Brodmann.area.9	0.011644241	162	6.120e−01	9.228e−01	MAS1:122 TMEM132D:141 NETO1:210 KCNC2:254 GPR26:260 KCNQ3:323
caudate.nucleus	−0.041313260	26	4.664e−01	9.228e−01	VAX1:141 ANO3:190 SLC35D3:926 GRM3:1776 SLC1A2:3005 FAM237A:3160
EBV.lymphocyte	−0.007665488	652	5.190e−01	9.228e−01	GPR183:89 NCKAP1L:93 SASH3:100 OASL:110 ANKRD44:147 HELB:164
ectocervix	−0.035713447	25	5.370e−01	9.228e−01	KCNK6:75 KRT34:811 PRLR:1184 MATN2:1347 GDF7:2315 TRHDE:4098
leg.skin	0.010699497	209	5.978e−01	9.228e−01	TRPM1:12 IL17RE:28 CDHR1:55 NKPD1:92 DSG3:102 FOXN1:145
lung	−0.017146748	139	4.883e−01	9.228e−01	CLEC5A:221 GPRC5D:441 ESM1:458 HGF:490 NDNF:774 ICAM1:974
minor.salivary.gland	−0.018944650	69	5.876e−01	9.228e−01	PROM1:359 CRACR2A:548 SERINC2:590 BPIFB6:683 ATP6V1B1:783 ELF5:938
substantia.nigra	−0.058919106	11	4.988e−01	9.228e−01	C10orf105:844 RET:1283 SLC6A3:2558 SPX:3208 TH:7018 KLHL1:7822
suprapubic.skin	−0.012702783	181	5.592e−01	9.228e−01	KRTDAP:15 CBLC:124 FAM25A:245 ANXA8L1:463 GRHL2:471 GAL3ST4:558
transformed.skin.fibroblast	−0.009478626	243	6.152e−01	9.228e−01	FKBP10:9 CYP7B1:22 FAP:69 KIRREL1:108 SLC38A2:240 LRP1:258
nucleus.accumbens	−0.013217643	93	6.610e−01	9.647e−01	VAX1:141 ANO3:190 DRD1:504 FAT3:538 NECAB2:910 COCH:947
left.ventricle	0.013340241	67	7.066e−01	9.784e−01	TNNI3K:46 XIRP1:826 LMOD3:878 CSRP3:968 MYOM2:1012 SMCO1:1127
tibial.artery	0.011013676	101	7.035e−01	9.784e−01	SUSD5:31 MN1:185 CRLF1:259 RERGL:306 SDK1:316 WFDC1:317
aorta	0.007035557	151	7.671e−01	9.863e−01	SUSD5:31 GABRR1:50 KCNK15:200 CNTN4:342 GAS6:366 HMCN1:389